

Assignment Title:

Global Optimization of Sourcing, Production, Workforce, and Logistics for Groupe Éléance

Module Code(s): IS6055

Module Title: Prescriptive analytics

Year: 2025-2026

Semester: 2

Due Date: **[6th April 2026]**

Abstract:

Groupe Éléance is a luxury perfume company headquartered in Grasse, France, with operations spanning procurement, production, warehousing, and international distribution. The company produces premium fragrances across three French factories (Grasse, Lyon, Bordeaux), uses two key warehouses (Paris and Marseille), and delivers globally to flagship retail cities: Paris, Milan, New York, Tokyo, Dubai, and Shanghai.

To meet growing international demand while aligning with sustainability targets, Groupe Éléance seeks to:

1. Optimize procurement of raw materials under supplier constraints (Table 1).
2. Allocate production across factories, considering costs, emissions, and labor availability (Table 2).
3. Plan staffing based on availability, absenteeism, and labor cost (Table 3).
4. Satisfy uncertain global demand from strategic cities, incorporating penalties and urgency levels (Table 4).
5. Store and dispatch from regional warehouses under capacity and carbon limits (Table 5).
6. Distribute products using multimodal transport networks with varying costs, speeds, and emissions (Tables 6, 7A, 7B).
7. Respect each product's material requirements for production (Table 8).
8. Apply a shortest-path algorithm (e.g., Dijkstra's) to determine optimal delivery routes from warehouses to cities via European and international hubs using the edge network (Table 9)

Your optimization model must account for competing goals:

- Minimize total cost (procurement, production, labor, storage, transport).
- Satisfy international demand under uncertainty and urgency.

- Reduce emissions from warehousing, transport, and production.
- Incorporate both continuous and integer variables.
- Implement network-based shortest-path modeling.

Assignment Tasks:

Part 1: Problem Analysis

- Identify operational and environmental risks (e.g., absenteeism, delivery variance, emissions).
- Discuss trade-offs between cost, time, emissions, and urgency.
- Highlight critical bottlenecks (e.g., factory capacity, supplier risk, warehouse constraints).

Part 2: Optimization Model Development

- You must incorporate CO₂ emissions into your model in one of two ways:
 1. Include total CO₂ emissions in the objective function using a cost-per-kg factor (e.g., €0.05 per kg CO₂), or
 2. Add a hard constraint limiting total emissions across procurement, production, storage, and transport (e.g., $\leq 500,000$ kg CO₂/month).
- Construct a transportation network from Table 9
- Use shortest-path algorithms (e.g., Dijkstra's) to:
 - Minimize delivery time for urgent shipments
 - Minimize delivery cost for non-urgent shipments
 - Compare alternative routing strategies
 - Identify risky or critical edges in the distribution network
- Define all decision variables:
 - Quantity purchased from each supplier
 - Units produced per factory per product
 - Staff assignment by role and location
 - Warehouse inventory levels
 - Quantity shipped from warehouse to city via transport mode

- Formulate the total cost objective function:
 - Include procurement, labor, production, warehousing, transport, and unmet demand penalties.
 - Minimize total cost, including:
 - Procurement
 - Production (fixed + variable cost)
 - Labor
 - Warehouse storage
 - Transportation
 - CO₂ emissions (if included in objective)
 - Penalties for unmet demand
- Apply constraints:
 - Supplier limits & delivery variance (Table 1)
 - Factory capacities & fixed cost (Table 2)
 - Workforce limits & absenteeism (Table 3)
 - Demand fulfillment under uncertainty (Table 4)
 - Warehouse storage and emissions (Table 5)
 - Feasibility of shipping based on mode and speed (Tables 6, 7B)
 - Material balance & product composition (Table 8)
 - Routing feasibility based on Table 9 edge travel times
- Apply advanced modeling techniques:
 - Mixed-integer linear programming (MILP)
 - Stochastic programming to model uncertainty in supply, demand, and labor

Part 3: Network Optimization and Simulation Analysis

- Construct a transportation network graph with warehouses and cities as nodes, and routes as edges weighted by travel time and/or cost.
- Apply shortest path or minimum-cost path logic to model time-based routing (e.g., for urgent deliveries).
- Run sensitivity analysis on:
 - Fuel cost fluctuations
 - Staff absenteeism rates

- Penalty cost changes
- Perform Monte Carlo simulation (500+ iterations) for uncertain parameters:
 - Supply variance
 - Demand variance
 - Absenteeism
- Evaluate scenario robustness and report findings.
- Run sensitivity analysis on:
 - Fuel cost fluctuations
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Part 4: Recommendations and Visualization

a. Present Actionable Recommendations

Prioritize Key Insights

Clearly highlight the most important outcomes from your model. Highlight:

- Best-performing suppliers under cost + CO₂ metrics
- Which factories to scale up/down
- Optimal mix of transport modes
- Most efficient warehouses for specific destinations
- Routing recommendations based on shortest-path analysis

For example:

- Which suppliers provide the best balance of cost, reliability, and CO₂ performance?
- Which factories should operate at full capacity, and which can be scaled down?
- Which shipping modes/routes minimize cost while meeting delivery deadlines?

- Which warehouses are most cost-effective for storing and dispatching products?

Propose Specific Actions

Provide data-driven operational strategies, such as:

- Increase procurement from Supplier X (e.g., EthanolPro) when costs are low and delivery risk is acceptable.
- Allocate high-volume production to Bordeaux to leverage lower CO₂ emissions per unit.
- Ship urgent orders via air from Paris and lower-priority orders via sea from Marseille.
- Assign additional bottling staff to Lyon to balance staffing shortages in Grasse.

b. Discuss Trade-Offs and Constraints

Analyze:

- Cost vs. emissions
- Speed vs. sustainability
- Supplier reliability vs. price
- Labor limits vs. production goals
- Warehouse capacity vs. deliveries
- How CO₂ caps alter the optimal solution

Capacity Limitations

- Identify bottlenecks: e.g., supplier caps (Table 1), labor shortages (Table 3), factory output (Table 2), or warehouse capacity (Table 5).
- Recommend reallocation strategies, e.g., increase staff at Lyon and shift production from Grasse during peak season.
- Model the cost and CO₂ implications of operating each factory at full vs. partial capacity.

c. Visualize Results Effectively

Graphs and Charts

- Heatmaps of transport cost & CO₂ by route
- Bar charts comparing factory production/emissions
- Pie charts of supplier contributions
- Network graphs showing:
 - Fastest paths

- Cheapest paths
- High-risk edges

Maps should show:

- Distribution routes
- Congestion or bottlenecks
- Warehouse loads

d. Provide Contingency Plans

Supplier Disruptions

- Model what happens if Supplier Capella (rare extracts) fails to deliver.
- Suggest substitutions: e.g., reduce production of Élégance A and increase B or C.
- Assess cost and CO₂ trade-offs when using alternate suppliers.

Plant Shutdowns

- Show reallocation logic if Bordeaux or Lyon shuts down temporarily.
- Quantify increase in cost and emissions under alternative production mix.

Demand Surges

- Recommend building buffer stock at Paris warehouse for high-urgency cities (e.g., New York, Tokyo).
- Consider using multimodal delivery (e.g., Road + Air) for flexibility.

e. Evaluate Sustainability Metrics

CO₂ Reduction Goals

- Compare total emissions in the optimized model vs. a baseline.
- Express improvement as a % reduction.

Energy Efficiency

- Show kWh/unit across factories.
- Recommend shifting more production to low-energy plants (e.g., Bordeaux).

Sustainability Alignment

- Explain how this optimization supports Groupe Élégance's net-zero ambitions.
- If emissions cap is modeled, show how the solution meets or exceeds targets.

f. Support Recommendations with Evidence

Statistical Summaries

- Total cost breakdown (procurement, labor, production, storage, transport, penalties)
- Fulfillment rates by city and by product
- Total CO₂ emissions across sourcing, production, storage, and transport

Scenario Analysis Results

- Sensitivity to:
 - Fuel price increase (10%, 20%)
 - Absenteeism shock (e.g., 20% bottling staff absent)
 - Surge in demand in NY or Shanghai

Supporting Data:

Table 1: Raw Material Suppliers

Supplier	Material	Cost/kg (€)	CO ₂ /kg	Max Supply	Lead Time	Variance
AromaVita	Essential Oils	42	1.5	12,000	5 days	±15%
LuxeCap	Bottles	8	0.7	40,000	2 days	±10%
EthanolPro	Ethanol	3	0.3	50,000	4 days	±20%
Capella	Rare Extracts	60	2.2	6,000	10 days	±25%

- This table outlines the available suppliers for each raw material, including unit costs, CO₂ emissions per kg, maximum monthly supply, lead time, and delivery variance.

Table 2: Production Facilities

Factory	Max Output	Fixed Cost	Cost/Unit	Energy Use	CO ₂ /unit	Staff/1k Units
Grasse	80,000	€50,000	€12	1.2 kWh	0.4 kg	7
Lyon	70,000	€45,000	€11	1.5 kWh	0.5 kg	6
Bordeaux	90,000	€60,000	€13	1.1 kWh	0.3 kg	8

- Details each factory's monthly output limit, fixed and variable production costs, energy usage, CO₂ emissions, and staff requirements per 1,000 units.

Table 3: Workforce

Role	Cost/Month	Max Staff	Absenteeism
Bottling	€2,500	40	±5%
Packaging	€2,800	35	±10%
QC	€3,000	25	±5%

- Lists available staff for each production role, monthly cost per staff member, and expected absenteeism rates.

Table 4: Retail Demand

City	Demand	Urgency	Penalty/Unit	Demand Variance
Paris	55,000	4	€5	±10%
Milan	35,000	3	€4	±12%
NY	65,000	5	€8	±20%
Tokyo	40,000	4	€6	±15%
Dubai	30,000	3	€5	±18%
Shanghai	50,000	4	€6	±15%

- Provides monthly product demand by city, urgency levels, penalty cost for unmet demand, and expected demand variability.

Table 5: Warehouses

Location	Capacity	Cost/Unit	CO2/unit	Distance to Factories (km)
Paris	100,000	€1.2	0.2 kg	Grasse: 900, Lyon: 600, Bordeaux: 500
Marseille	120,000	€1.0	0.3 kg	Grasse: 200, Lyon: 400, Bordeaux: 600

- Includes warehouse capacities, storage costs, CO₂ emissions per stored unit, and distance from each factory.

Table 6: Transport Modes

Mode	Cost/km	CO2/km	Speed	Reliability
Air	€0.08	0.025	900 km/h	3
Sea	€0.03	0.010	30 km/h	4
Rail	€0.04	0.012	60 km/h	5
Road	€0.05	0.015	80 km/h	3

- Provides cost and emission rates per km per unit, average speed, and reliability rating for each transport mode (air, sea, rail, road).

Table 7A: Distance Matrix (Warehouse → City in km)

From \ To	Paris	Milan	NY	Tokyo	Dubai	Shanghai
Paris	0	850	5,800	9,700	5,200	9,300
Marseille	800	1,050	6,000	9,900	5,400	9,400

- Lists distances (in km) from each warehouse to each destination city.

Table 7B: Travel Time Matrix (hours)

From \ To	Paris	Milan	NY	Tokyo	Dubai	Shanghai
Paris	0	10.6	6.4	10.8	5.8	10.3
Marseille	10	13.1	200	330	180	313

- Lists estimated delivery times (in hours) for each route, depending on transport mode.

Table 8: Product Composition (per unit)

Product	Oils (g)	Ethanol (ml)	Bottles	Extracts (g)
A	5	50	1	2
B	4	60	1	1
C	3	45	1	0.5

- Specifies how much of each raw material is needed to produce one unit of each product (A, B, C).

Table 9: Transport Network Edge List — Warehouse to City via Hubs

From Node	To Node	Transport Mode	Distance (km)	Travel Time (hrs)	Notes
W1: Paris WH	E1: Lyon	Road	450	5.6	Fastest road to SE hub
W1: Paris WH	E3: Frankfurt	Road	480	6.0	High-speed road link
W1: Paris WH	E4: Amsterdam	Road	350	4.4	Northern EU routing
W2: Marseille WH	E1: Lyon	Road	300	3.8	Regional supply path
W2: Marseille WH	E2: Milan	Road	800	10.0	Direct to city
W2: Marseille WH	E3: Frankfurt	Road	900	11.3	Alt. to Amsterdam
E1: Lyon	G2: Frankfurt Airport	Road/Air	400	5.0	Routing via FRA hub

E2: Milan	G2: Frankfurt Airport	Road/Air	300	3.8	Via central Europe
E3: Frankfurt	G1: Paris CDG	Road/Air	500	6.3	Fast link to CDG
E4: Amsterdam	G1: Paris CDG	Road/Air	430	5.4	Northern access route
G1: CDG	D3: New York	Air	5800	6.4	Long-haul route
G1: CDG	D4: Tokyo	Air	9700	10.8	Long-haul route
G1: CDG	D5: Dubai	Air	5200	5.8	Mideast via CDG
G1: CDG	D6: Shanghai	Air	9300	10.3	Asia via CDG
G2: FRA	D3: New York	Air	6000	6.7	Alt. routing via FRA
G2: FRA	D6: Shanghai	Air	9400	10.4	Central EU → China
G3: Dubai Port	D4: Tokyo	Air	8000	8.9	Regional direct link
G3: Dubai Port	D6: Shanghai	Air	6000	6.7	Mideast → East Asia
G4: Singapore	D4: Tokyo	Air	6200	7.2	SEA hub → Japan
G4: Singapore	D6: Shanghai	Air	5000	5.6	SEA hub → China
W2: Marseille WH	G3: Dubai Port	Sea	5400	180.0	Long-haul sea freight
W2: Marseille WH	G4: Singapore	Sea	9000	300.0	Long-haul sea freight
W2: Marseille WH	G5: Shanghai Port	Sea	8800	293.0	SEA shipping line

- This table lists all transport edges between warehouses, hubs, and retail cities, including transport modes and estimated travel time in hours.

- Quick Reference:**

Code	Meaning	Node Label
CDG	Charles de Gaulle Airport	G1
FRA	Frankfurt Airport	G2
DXB	Dubai Airport/Port	G3
SIN	Singapore (Changi)	G4

Note:

- **These tasks must be implemented using Python. You need to submit your Python code along with your report.**

Required Format:

- a) 6,000 words excluding references and statistical summaries or programming codes.
 - b) All external sources of information must be referenced.
 - c) Font – Times New Roman size 12.
 - d) Spacing – 1.5.
- **Referencing Style:** For this course, you should use the **Harvard referencing style**. This includes both in-text citations and a reference list at the end of your assignment.
 - **In-Text Citations:** When you refer to or paraphrase someone else's work within your text, you must include an in-text citation. Here are a couple of examples:
 - **Book:** (Ladley, 2020)
 - **Journal Article:** (Khatri & Brown, 2010)
 - **Website:** (Data Governance Institute, 2023)
 - **Reference List:** At the end of your assignment, include a full list of references with all the details of each source. Examples:
 - **Book:** Ladley, J. (2020) *Data Governance: How to Design, Deploy, and Sustain an Effective Data Governance Program*. 2nd ed. Waltham: Elsevier.
 - **Journal Article:** Khatri, V. and Brown, C.V., 2010. Designing data governance. *Communications of the ACM*, 53(1), pp.148-152.
 - **Website:** Data Governance Institute (2023) *The Data Governance Framework*. Available at: <https://www.datagovernance.com/framework> (Accessed: 15 August 2024).

Group Formation Instructions

- **Group Size:**
Each group must consist of **6 or 7** members.
- **Group Roles:**
Each group member should take responsibility for a specific aspect of the assignment.
- Schedule regular meetings to discuss progress and resolve challenges.
- Use collaborative tools like Microsoft Teams to work together efficiently.
- Create a timeline with milestones for each part of the assignment.
- If you are unable to find a group, please notify me by **the end of Week 3**. You will be assigned to a group based on availability.

Submission & Formatting Guidelines:

Due Date: [6th April 2026]

When submitting your assignment, please ensure that your file is properly formatted and named according to the following guidelines:

- **File Name Format:**

- IS6055-Group Assignment-Your group number
 - Example: IS6055- Group Assignment-Group1

- **File Type:**

- Please submit your work as a **PDF** or **Word document (.docx)** along with the **Python file** (PyCharm or Jupyter)

- **File Content:**

- **Title Page:** Include a title page with the following information:
 - Module Code: IS6055
 - Module Title: Prescriptive Analytics
 - Group Number (e.g., IS6055-Group1)
 - Full Names and Student IDs of all members
 - Primary Contact for the group (email/phone)
 - Submission Date

Footers: Include page numbers in the footer of each page.

- **Submit your assignment as a Word/PDF document along with the Python file (PyCharm or Jupyter) through Canvas.**
- **All assignments will be checked for plagiarism using Turnitin.**
- **All assignments will be graded according to the rubric provided in Table-Rubric.**

Assessment Criteria:

Table Rubric: Assessment Criteria for Groupe Éléance Optimization Assignment

	Criteria	Excellent (85–100%)	Good (70–84%)	Satisfactory (50–69%)	Needs Improvement (30–49%)	Unsatisfactory (0–29%)
Part1	Problem Analysis & Trade-Offs (15%)	Thorough, insightful analysis with clear identification of risks, trade-offs, and bottlenecks.	Good analysis with minor gaps in identifying bottlenecks or trade-offs.	Adequate analysis but lacking depth or missed key interactions.	Weak analysis with unclear problem framing.	Minimal or no meaningful analysis.
Part2	Optimization Model Development (30%)	Clearly defined variables, objective, constraints, and integration of all cost areas; emissions handled well.	Mostly complete model with minor omissions or unclear logic in some areas.	Basic model structure with errors in constraints or missing some components.	Poorly formulated model with missing logic or incorrect constraints.	No working model or major structural flaws.
Part3 (20%)	Network & Shortest Path Logic	Accurately models transport network using cost/time; clear justification of mode selection.	Valid network model with some limitations or assumptions unexplained.	Simplified routing logic; minor misalignment with data.	Incorrect or vague network setup; minimal use of routing principles.	No attempt to model network logic or paths.
	Stochastic Programming	Excellent integration of uncertainty in supply/demand/labor; well-justified scenarios.	Good use of stochastic logic; some simplification or unsupported assumptions.	Basic scenarios modeled but lack integration or clarity.	Attempted uncertainty modeling but unclear or incorrect.	No treatment of uncertainty or stochastic input.
	Monte Carlo & Sensitivity	Simulation well executed; explains variation & response; sensitivity clearly demonstrated.	Mostly complete simulation with adequate interpretation.	Some useful sensitivity/scenario outputs but unclear impact analysis.	Simulation results weak or unconnected to decision-making.	No scenario or simulation modeling attempted.
Part4	Recommendations & Visualization (25%)	Actionable, well-justified recommendations; strong visuals, CO ₂ metrics, and trade-offs addressed.	Clear insights with effective graphs; minor issues in depth or presentation.	General recommendations with basic visuals and CO ₂ coverage.	Limited or unsupported recommendations; weak visuals.	No clear recommendations or visualizations.
Formatting & Referencing	Formatting & Referencing (5%)	All formatting correct; full, consistent Harvard references throughout.	Minor formatting or referencing issues.	Noticeable formatting or citation errors.	Poor formatting and inconsistent references.	No referencing or formatting observed.
AI	AI Supervisory Skills (5%)	Demonstrates exceptional ability to supervise AI tools effectively. Critically evaluates and refines AI-generated content. Clear evidence of supervision and	Shows good ability to supervise AI tools. Evaluates AI output but may miss minor issues. Evidence of supervision and decision-making is	Demonstrates basic AI supervision skills. Occasionally accepts AI output without critical evaluation. Evidence of	Shows limited ability to supervise AI tools. Often accepts AI output without critical evaluation. Evidence of supervision is	Fails to supervise AI tools effectively. Blindly accepts AI output without evaluation. Evidence of supervision is absent or unconvincing.

		decision-making in reports and code.	present but could be more detailed.	supervision is minimal or unclear.	vague or insufficient.	
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AI Supervisory Skills:

Note: If you choose to use **AI tools like ChatGPT** to assist with your assignment, you must provide transparency in your work. Specifically, if you incorporate any content generated by ChatGPT or similar AI tools, you are required to share the link to the specific conversation or session with me along with your assignment submission. This ensures that all sources of information are properly acknowledged, and it helps maintain the integrity of your work. Failure to disclose the use of such tools may be considered a breach of academic honesty policies.

- As you interact with AI tools like ChatGPT, your task is not only to use the tool but to critically supervise the suggestions it provides. This means evaluating the accuracy, relevance, and applicability of AI-generated content. You should refine and adapt the AI's suggestions to better suit your assignment's context.
- You are required to share the code and the final report with clear documentation on how you used AI assistance. Your documentation should explain how you supervised the AI, what decisions you made to modify or accept its suggestions, and why.
- Your ability to supervise the AI will be evaluated based on how well you refine and adapt the AI-generated content. You will be assessed on your critical thinking, decision-making, and the quality of your final outputs. The emphasis will be on ensuring that you use AI as a tool to enhance your work, not as a crutch to produce uncritical or unrefined results.

Feedback Mechanisms

INTERIM FEEDBACK: Project groups may request feedback on progress throughout the course of the project via the following mechanisms: (1) Scheduled meeting with lecturer or (2) Email feedback on draft submissions.

Penalties

Submission of a project indicates that this is the student's own work. If the submission is found not to be original work, a mark of 0% will be automatically awarded. This project is worth 50% of the total module percentage for IS6055. Where work is submitted up to and including 7 days late, 10% of the total marks available shall be deducted from the mark achieved. Where work is submitted up to and including 14 days late, 20% of the total marks available shall be deducted from the mark achieved. Work submitted 15 days late or more shall not be accepted.

This project outline adheres (as appropriate) to the policies and codes of conduct outlined in Handbook Governing Curriculum Approval, the Policies and Guidelines Governing Academic Programmes, DMIS Guidelines, the Guide to Examinations for Staff and Students, the Plagiarism Policy, the Duty of Respect and Right to Dignity Policy, and the UCC Student Rules etc.